common method of leaving the present vehicle. To be sure, packs may bud off daughter packs. But the genes in the parent pack don't pass to the daughter pack in a single vessel in which all have an equal share. The genes in a pack of wolves don't all stand to gain from the same set of events in the future. A gene can foster its own future welfare by favouring its own individual wolf, at the expense of other individual wolves. An individual wolf, therefore, is a vehicle worthy of the name. A pack of wolves is not. Genetically speaking, the reason for this is that all the cells except the sex cells in a wolf's body have the same genes, while, as for the sex cells, all the genes have an equal chance of being in each one of them. But the cells in a *pack* of wolves do not have the same genes, nor do they have the same chance of being in the cells of sub-packs that are budded off. They have everything to gain by struggling against rivals in other wolf bodies (although the fact that a wolf pack is likely to be a kin group will mitigate the struggle).

The essential quality that an entity needs, if it is to become an effective gene vehicle, is this. It must have an impartial exit channel into the future, for all the genes inside it. This is true of an individual wolf. The channel is the thin stream of sperms, or eggs, which it manufactures by meiosis. It is not true of the pack of wolves. Genes have something to gain from selfishly promoting the welfare of their own individual bodies, at the expense of other genes in the wolf pack. A beehive, when it swarms, appears to reproduce by broad-fronted budding, like a wolf pack. But if we look more carefully we find that, as far as the genes are concerned, their destiny is largely shared. The future of the genes in the swarm is, at least to a large extent, lodged in the ovaries of one queen. This is why—it is just another way of expressing the message of earlier chapters—the bee colony looks and behaves like a truly integrated single vehicle.

Everywhere we find that life, as a matter of fact, is bundled into discrete, individually purposeful vehicles like wolves and bee-hives. But the doctrine of the extended phenotype has taught us that it needn't have been so. Fundamentally, all that we have a right to expect from our theory is a battleground of replicators, jostling, jockeying, fighting for a future in the genetic hereafter. The weapons in the fight are phenotypic effects, initially direct chemical effects in cells but eventually feathers and fangs and even more remote effects. It undeniably happens to be the case that these phenotypic effects have largely become bundled up into